

Supervised Fine-Tuning

DATA 443/643 · Week 8, note 8.1

Week 8

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Ideas from Stanford CS224N 2025 L10 (instruction finetuning as InstructGPT stage 1). Original slides; that course is not copied.

Lecture plan

- 1 Pretrain next-token, then follow a request
- 2 Mask the prompt
- 3 Quality beats noisy volume
- 4 The template is part of the method

Key goal: SFT copies answers. Preferences are Week 9.

Same transformer, different data

- The base model already scores every continuation.
- Instruction fine-tuning is the first InstructGPT stage.
- Alpaca and Vicuna are this stage with public pairs.

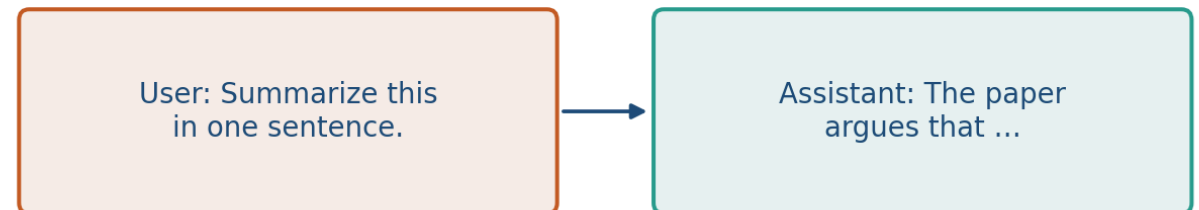
Instruction data is a distribution shift, not a new architecture



A row you would imitate

- Not a Wikipedia paragraph.
- A task plus an answer you would accept.
- System prompts and schemas live in the prompt, not a new net.

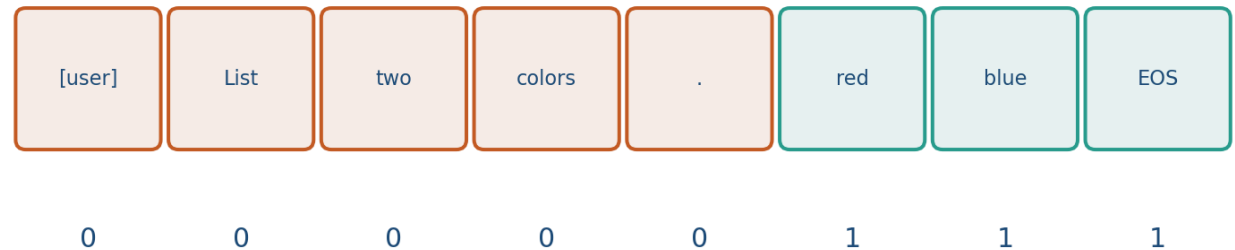
One SFT row: a request and a completion you would actually want



Zeros on the user

- Forward the prompt. Zero those positions in the loss.
- Otherwise the model spends gradient on imitating the user.
- Ten-token answers and 400-token answers are different datasets.

SFT mask: zeros on the prompt, ones on the response.



What should dominate the first run

80 clean medical pairs

Lead. Every response token is a label you want.

80,000 noisy threads

Token average lets the forum win 10:1. Filter or down-weight.

What to take from this lecture

Mask

Loss on Tr response tokens, not Tp + Tr.

Template

Train and eval with the same roles.

Not RLHF

SFT copies. Chosen-versus-rejected is Week 9.